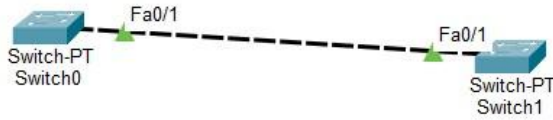


Vlan trunking and access port theory

1. In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol. ●



Switch0

IOS Command Line Interface

```
%SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1

Switch#
```

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>
Switch>ena
Switch>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f0/0
%Invalid interface type and number
Switch(config)#int f0/1
Switch(config-if)#switch
Switch(config-if)#swit
Switch(config-if)#switchport mode trunk
Switch(config-if)#swi
Switch(config-if)#switchport trunk encapsulation dot1q

% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

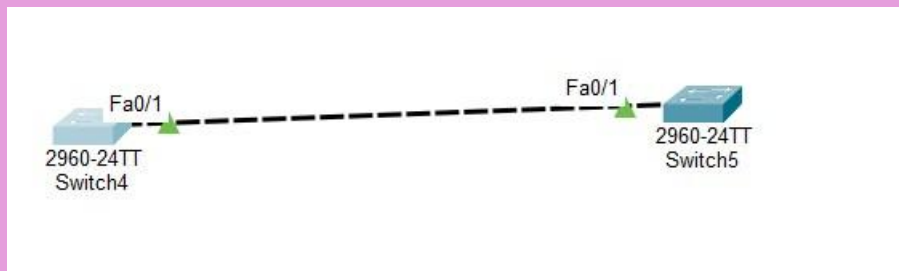
Port      Vlans allowed and active in management domain
Fa0/1     1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1

Switch#switchport trunk encapsulation dot1q
```

Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

2.



```
Switch>
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#int f0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int f0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

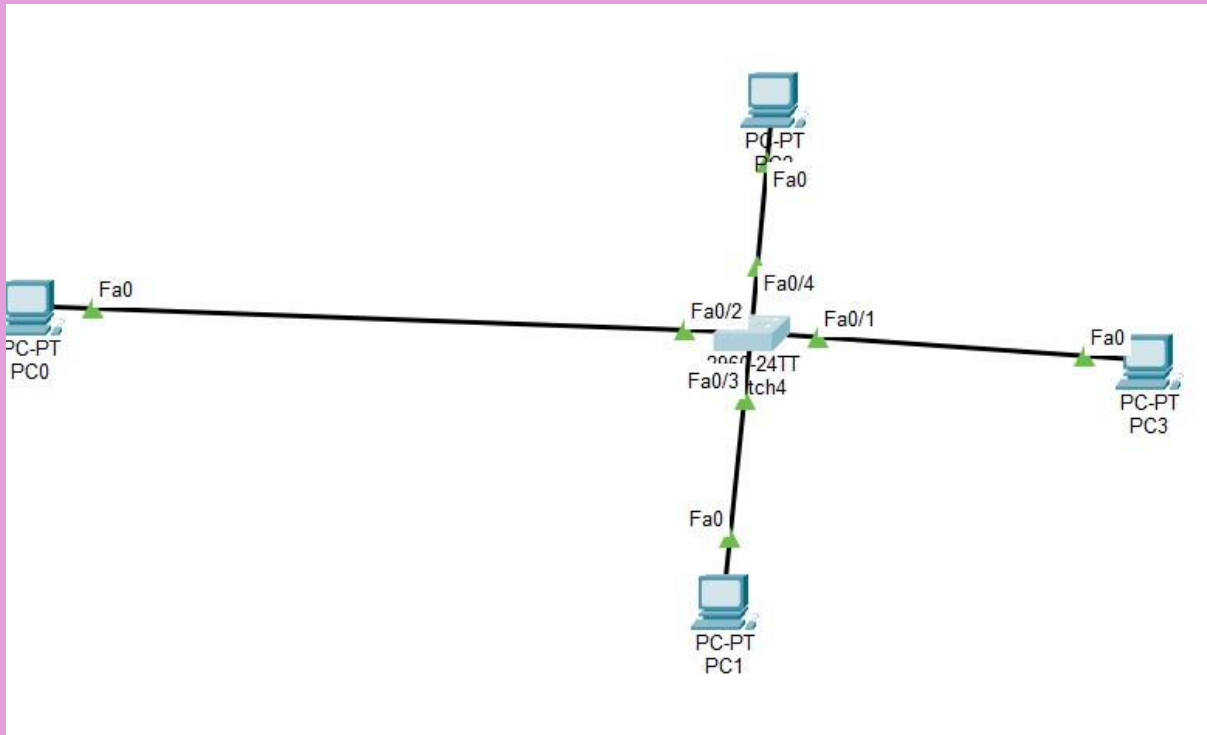
```
Switch#
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/2
20	VLAN0020	active	Fa0/3
30	VLAN0030	active	Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN. ●

3.



```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

Hint: Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.

4.



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End
with CNTL/Z.
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
-----
Fa0/1     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1
```

5. Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.



❏ Access vs Trunk Behavior

- Access port → sends/receives frames untagged, belongs to a single VLAN (here VLAN 20).
- Trunk port → expects tagged frames for multiple VLANs, used between switches/routers.

❏ What Happens to the PC

- PCs do not tag frames; they send plain Ethernet frames.

- The trunk port expects VLAN tags, so it drops or misinterprets the PC's traffic.
- Result → PC cannot communicate with other VLAN 20 devices. Ping/ARP requests fail.

• Symptoms Observed

- PC shows “Request Timed Out” when pinging VLAN 20 hosts.
- Switch logs may show input errors or unrecognized frames.
- Other VLANs remain unaffected, only this PC loses connectivity.

Corrective Action

- Reconfigure FastEthernet0/5 back to an access port:

```
Switch2> enable
Switch2# configure terminal
Switch2(config)# interface fa0/5
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 20
```